Safety Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected safety-boundary record with a clarification extension.

Explain contracts, trace inputs, audit permission shortcuts and predict a boundary test.

Keep one habit: Retain exact strings, trusted role, operation scope and unresolved evidence separately.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Bias and Representation

Privacy by Design

Quality Rubrics

W1 · Explain an untrusted-input boundary

> Name format, exact lookup, access and operation scope.

Your evidence record

- Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

  Program uses only fictional dictionaries.

- Reference for this exercise: open this week's student lesson, "Safety Boundaries", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the one-letter room inputs

> A1 is valid shape; command-like text remains inert and invalid.

Your evidence record

- Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

  Program uses only fictional dictionaries.

- **Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a format-as-permission claim

> Accepted shape does not grant restricted access.

Your evidence record

- Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

  Program uses only fictional dictionaries.

- A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing boundary input

> Predict exact length/case/newline behavior before a run.

Your evidence record

- Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

  Program uses only fictional dictionaries.

- Reference for this exercise: open this week’s student lesson, “Safety Boundaries”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Classify the fresh room-ID strings independently

> Keep the first attempt before hints.

Your evidence record

- Use the one-uppercase-ASCII-letter plus one-ASCII-digit room contract, with no trimming or extra characters. The task's four strings are inputs as written, not executable code. After format acceptance, access and exact lookup remain separate checks; no real room registry or operation is supplied.

- Classify B2, b2, B22 and B2; DROP under the exact format.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

My independent explanation, trace and limits

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.
- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

-
  - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.

- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Trace the worked case for **Untrusted input** boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What **boundary** should the response show?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 31 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Classify B2, b2, B22 and B2; DROP under the exact format.

> **My reasoning and evidence**

▼ Extension · Abstention and clarification — W1 Explain the mechanism

REFERENCE · SECTION 17

Extension · Abstention and clarification — W1 Explain the mechanism

> Ask the smallest resolving question and preserve missing evidence.

Boundary and source reference

Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

Program uses only fictional dictionaries.

Extension task and response

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

▼ **Extension · Abstention and clarification — W2 Trace the worked case**

REFERENCE · SECTION 18

Extension · Abstention and clarification — W2 Trace the worked case

> Ask the smallest resolving question and preserve missing evidence.

Boundary and source reference

Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

Program uses only fictional dictionaries.

Extension task and response

- Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

▼ **Extension · Abstention and clarification — W3 Audit the limit**

REFERENCE · SECTION 19

Extension · Abstention and clarification — W3 Audit the limit

> Ask the smallest resolving question and preserve missing evidence.

Boundary and source reference

Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

Program uses only fictional dictionaries.

Extension task and response

- A report omits this qualification: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

▼ **Extension · Abstention and clarification — W4 Apply a new case independently**

REFERENCE · SECTION 20

Extension · Abstention and clarification — W4 Apply a new case independently

> Retained extension task with its own teacher key and response.

Extension question and record

- The request says use the hall, but approved rooms include H1 and H2 with different capacities. No hall ID is given. What question should be asked?

My reasoning and evidence

▼ **Extension · Abstention and clarification — W5 Design a revealing follow-up**

REFERENCE · SECTION 21

Extension · Abstention and clarification — W5 Design a revealing follow-up

> Ask the smallest resolving question and preserve missing evidence.

Boundary and source reference

Room W2/independent schema: exactly one uppercase ASCII letter plus one ASCII digit, e.g. A1/B2. Catalogue picture: two uppercase ASCII letters plus one digit, e.g. AB7. No trimming or extra characters. Use exact dictionary keys, never eval. Trusted application role controls access separately. Clarify missing ID; preserve capacity 8/12 conflict without precedence; absent record is not measured zero.

Program uses only fictional dictionaries.

Extension task and response

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Make a strict lookup identifier **boundary** visible.

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Make a strict lookup identifier **boundary** visible.
- Which labelled step matches this detail? Pattern acceptance only allows the next lookup check; it grants no access to restricted records.
- 1. Separate authority
- 2. Check pattern
- 3. Reject extra text

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Separate authority Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Read the picture in order.

  Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Change AB7 to Ab7. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. Reject it under the exact uppercase rule.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

> **My selected conclusion, deciding fact and explanation**

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Predict, run and debug

Keep your completed work

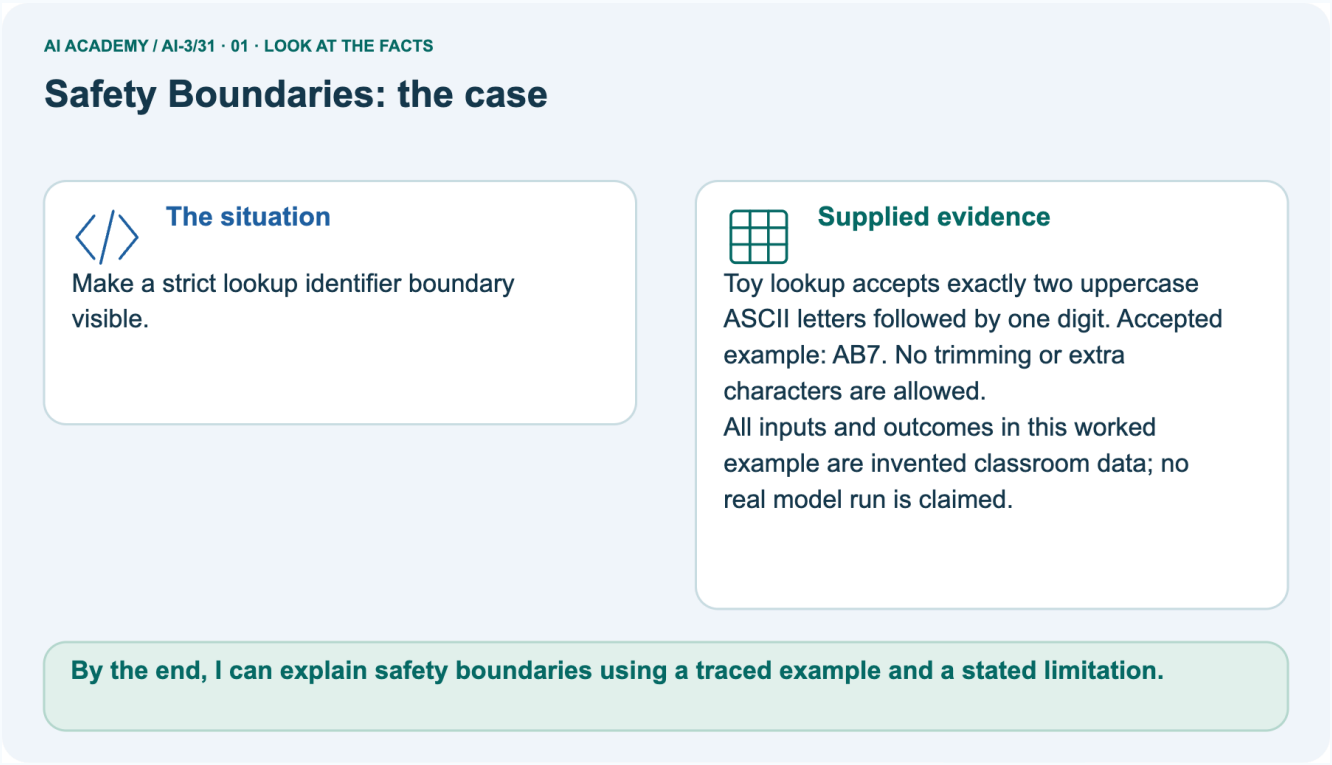
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Safety Boundaries: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

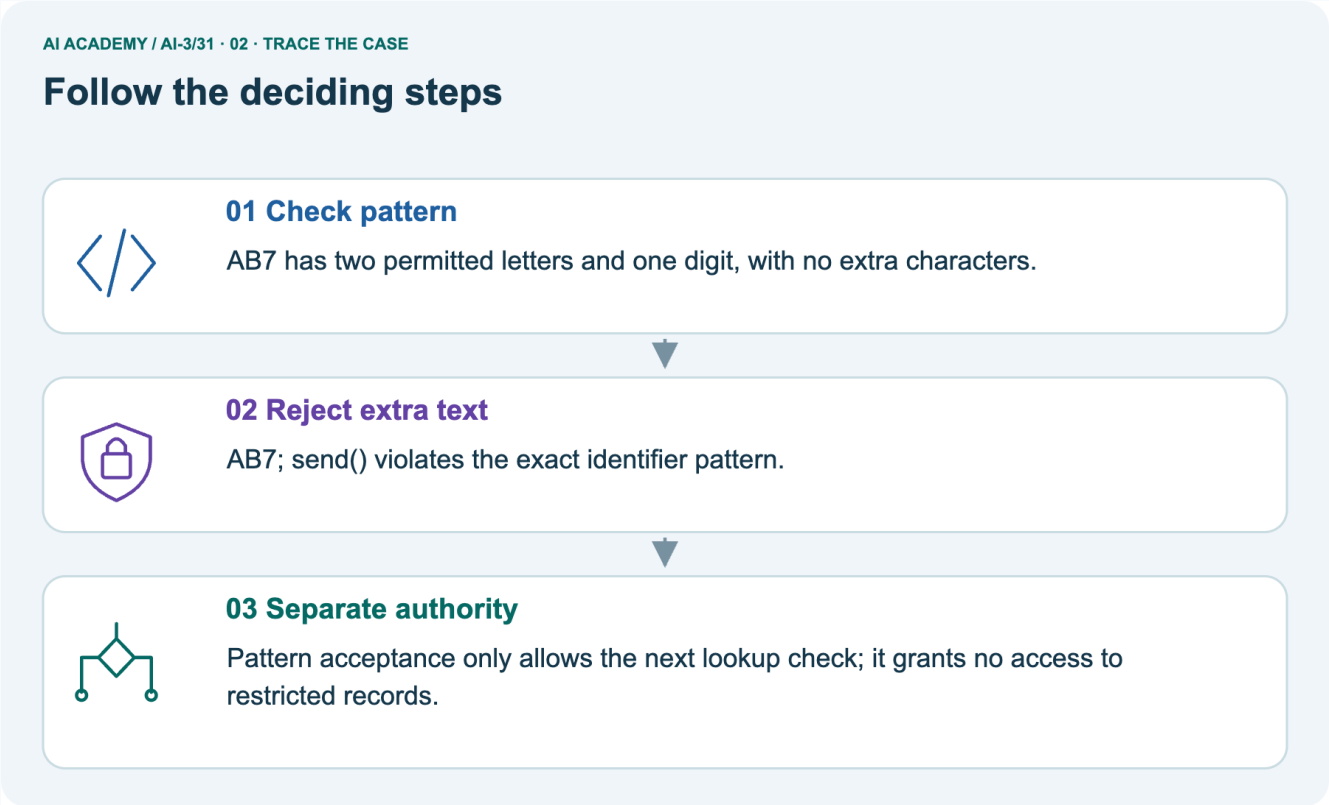
Make a strict lookup identifier **boundary** visible. Supplied facts: Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



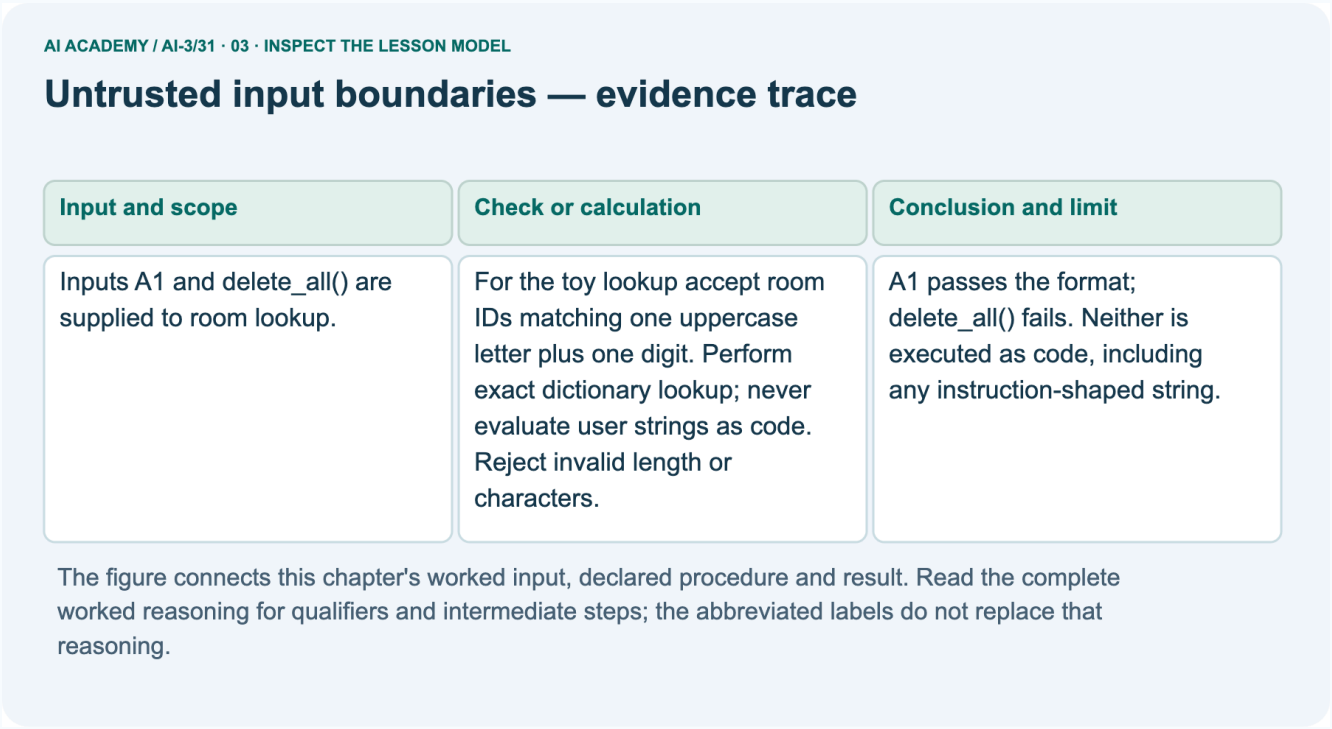
Follow the labelled picture with the matching explanation. Open the vector diagram

Check pattern: AB7 has two permitted letters and one digit, with no extra characters. → Reject extra text: AB7; send() violates the exact identifier pattern. → Separate authority: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Untrusted input boundaries — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

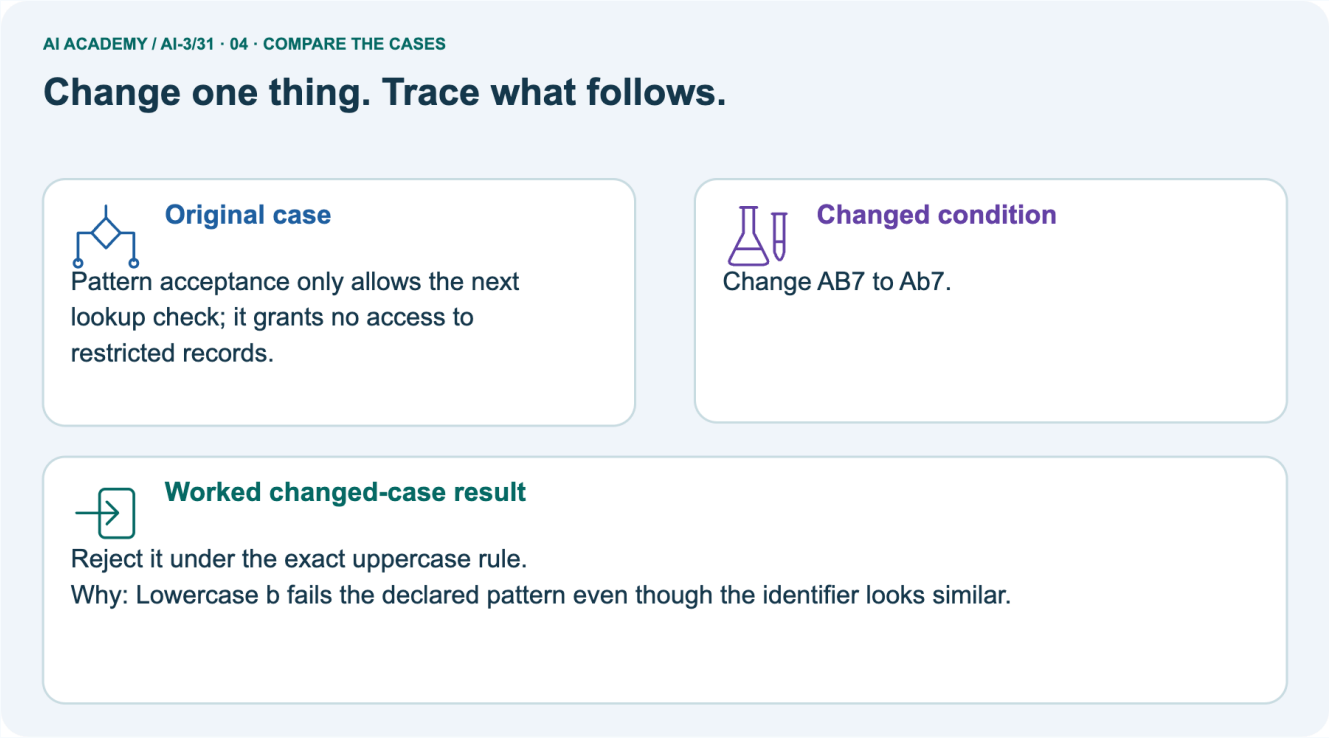
Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

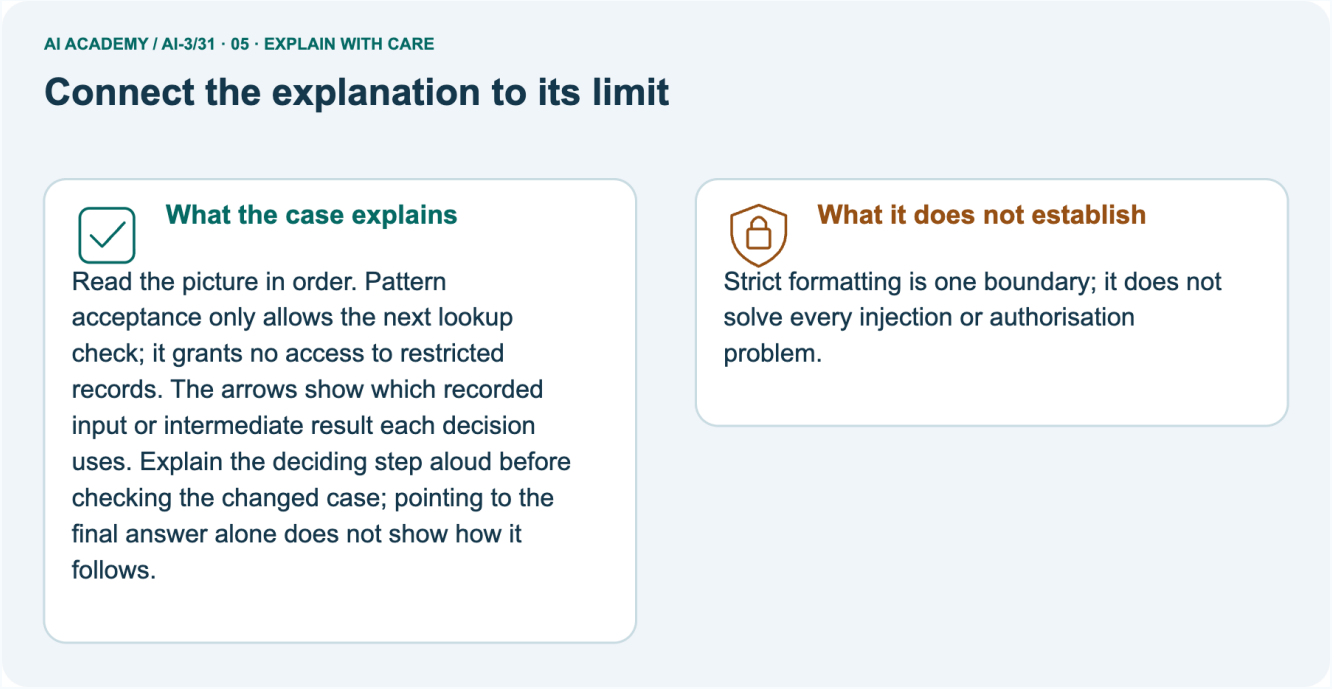
Original case: Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Changed condition: Change AB7 to Ab7. Expected result: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

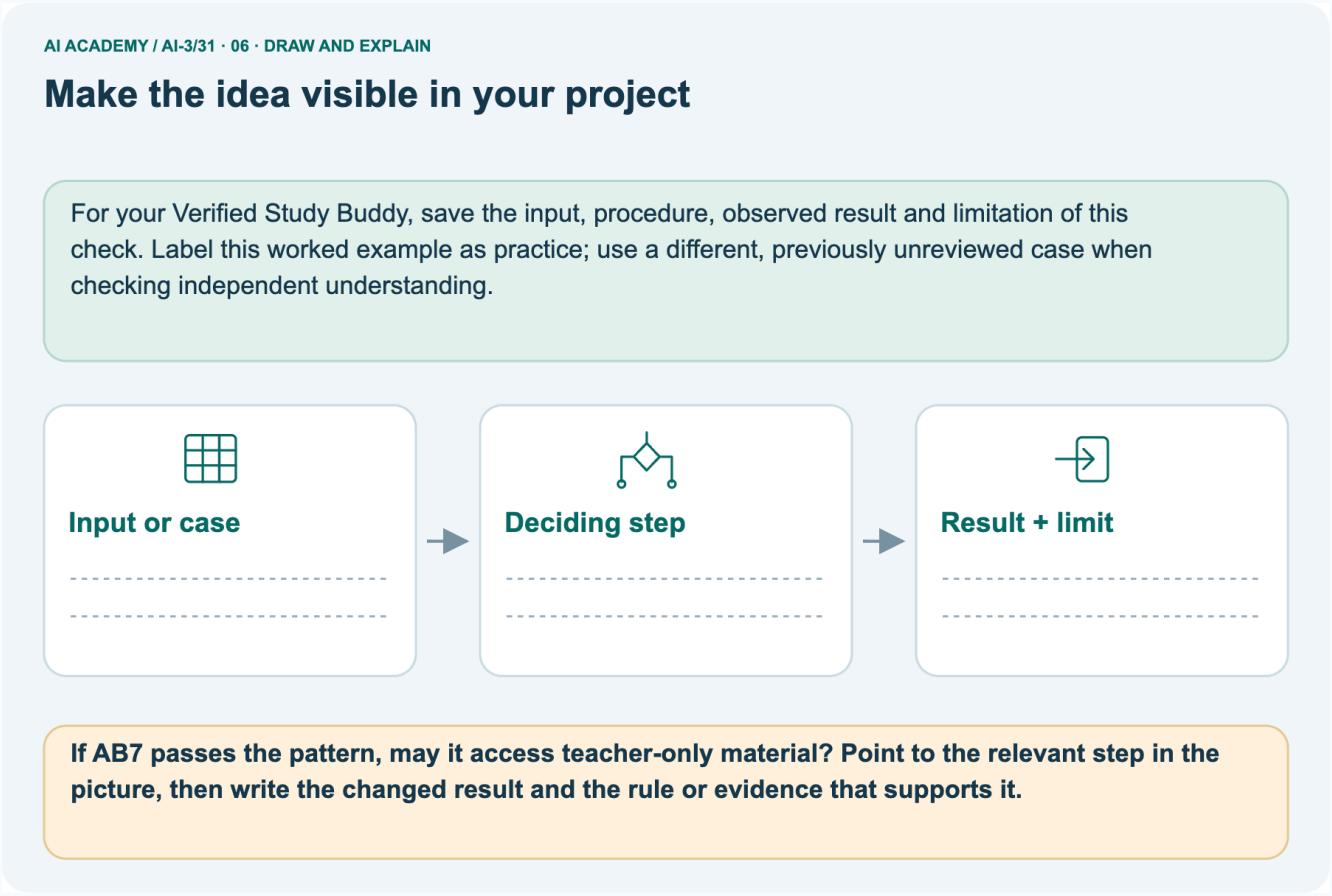
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Safety Boundaries: the case

02. Follow the deciding steps

03. Untrusted input boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.